

Continuous Distillation

Experiment 4
CHE382: Unit Operations Laboratory II
Tuesday 9 AM
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Group 1:
James Fell
Diego Martinez
Allison Nordvall
William Rawson
Emily Sobie
Peter Starakiewicz

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Executive Summary

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The overall objective of the *Continuous Distillation* lab is to study the characteristics of a continuous distillation column when varying the feed location. This experiment is split into three parts. The objective of part A is to determine the calibration curve for the pump, as well as to set up an accurate flowrate of feed pumped into the column. The objective for parts B and C is to observe effects on the distillation while the feed location is changed. The effects can be analyzed by determining the number of theoretical stages and by determining the overall efficiency of the column.

The results of the Continuous Distillation experiment show that the location of the feed affects the purity of the distillate and the temperature profile of the column. In part A, the goal was to determine if the pump was accurate. When the speed of the pump was increased, there was a linear increase in the flowrate. The R^2 value for the pump calibration curve was 0.998 which tells us that the data follows a strong linear correlation. In part B, the feed was placed between trays 4 and 5. While the feed stream was in the middle of the column, the results show that the concentration of ethanol in the top decreased, and the concentration of ethanol in the bottom increased. For example, the first sample taken from the top of the column had an ethanol concentration of 99.19 wt%, and the last sample from the top had an ethanol concentration of 98.48 wt%. The first sample taken from the bottom of the column had a concentration of 8.77 wt%, and the last sample from the bottom had an ethanol concentration of 10.77 wt%. In part C, the feed stream was placed at the top of the column. Like part B, the concentration of ethanol in the top decreased as the experiment progressed, while the concentration for ethanol in the bottom increased. For example, the first sample that was taken from the top of the column had an ethanol concentration of 98.29 w%, and the last sample from the top had an ethanol concentration of 97.27 wt%. The first sample taken from the bottom of the column had a concentration of 12.69 wt%, and the last sample from the bottom had an ethanol concentration of 15.69 wt%. Overall, the concentration of ethanol changed as the feed was positioned higher on the column. There was an average of 99% concentration of ethanol when the position of the feed was in the middle and 98% at the top.

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Overall, the Continuous Distillation experiment showed that the placement of the feed directly affects the purity of the products. Our results show a decrease of ethanol concentration of in the distillate when the feed was located at the top of the column, and an increase when it was located between stages 4 and 5. This means that the lower the feed is placed in the column, the more pure the distillate product will be.

Introduction

In this experiment, a binary mixture of ethanol and water was separated by continuous distillation, to demonstrate the separation effects based on the feed location. The equipment used in this experiment was the Armfield UOP3CC Continuous Distillation Column, it was controlled using a bench-mounted control console. A computer connected to the console was used to gather all the data during the experiment. The reboiler contained a 10 L mixture made-up of 10%wt ethanol and water. This was used to start-up the column and heat the feed stream. A separate 3 L mixture of 30%wt of ethanol and water was used as the feed solution. The first part of the experiment was to make a calibration graph of the actual flowrate vs the position of the variable speed dial on the control console. This was done by disconnecting the feed tube from the distillation column and then turning on the pump and purging the system before taking any measurements. When the flow of the liquid exiting the feed tube was constant, the flow rate was measured using a graduated cylinder and a stopwatch. This procedure was used for the speed dials 1-10. The second part of the experiment was continuous distillation with the feed tube located between the 4th and 5th trays. To begin, the power of the reboiler was set to 1.5 kW to reach the columns operating temperature. When enough ethanol started to build up in the distillate receiver, the power was set to 0.75 kW and reflux was set to total reflux. When a constant temperature was observed, it was determined that the column was stable. After the column stabilized, reflux of 5:1 was set, and the heat exchanger was turned on. The flow rate of the feed stream was set to 30-35 mL/min. This rate was determined by using the calibration curve of the knob location to flowrate. After feeding about 500 mL of the 3L mixture, a sample of the distillate and bottoms were taken simultaneously. Once the initial samples were collected, more were obtained after waiting for five-minutes per each pair of samples. Once the samples were chilled to room temperature, the refractive index of each sample was measured. The same procedure performed in part 2 was also performed for part 3. The only difference between experiments 2 and 3 is that the feed location of the third experiment was located at the top of the column. Distillation is widely used in numerous chemical industries because it provides a simple method of separating a liquid-vapor mixture of two or more components.

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Theory

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The theory of continuous distillation is to effectively separate the more volatile components from the less volatile components in a feed stream. The more volatile components will be the distillate stream leaving the top of the column and the less volatile components will be the bottoms stream leaving the bottom of the column. The feed stream is conventionally placed near the middle column. However, depending on the purpose of the distillation column, the feed location can be changed. If the distillation column is used as a stripping column, the feed location is placed more towards the top of the column. Conversely, if the distillation column is used as a rectifying column, the feed stream is placed more towards the bottom of the column. Once the feed stream enters the column, the components are separated by their respective boiling points.

The components of the feed with the lower boiling points will become the distillate leaving the top of the tower, and the components of the feed with higher boiling points will become the bottoms leaving the bottom of the tower. The volatilities of the components in the feed stream also play an important role in effectively separating the feed stream into high purity product streams. To determine the relative volatility of the feed mixture, ethanol and water in this case, the following equation can be used [4]:

$$\alpha = \frac{(y_{EtOH}/x_{EtOH})}{(y_{H_2O}/x_{H_2O})} \quad (1)$$

Where x_i are the liquid mole compositions and y_i are the vapor compositions.

Vapor formed in the column will travel upstream to the top of the column. Conversely, the liquid present in the column will travel downstream. Every distillation column has stages where the vapor and liquid come in contact with each other. If the distillation column is operating ideally, there will be vapor-liquid equilibrium present at each stage in the column. The reflux ratio is a crucial variable regarding the distillation. The reflux ratio has an effect on how pure the product stream, the distillate, is. In the case of continuous distillation, the reflux ratio used will remain constant throughout operation. [1] Near the bottom of the column is the reboiler, which supplies the heat needed to boil the feed mixture. The vapor product leaving the top of the column is condensed. Figure 1 below shows a schematic of a distillation column.

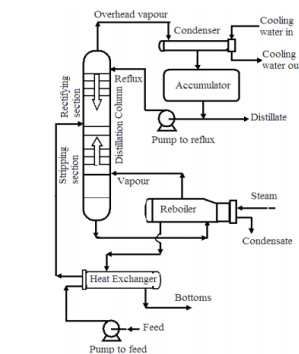


Figure 1: Schematic of a distillation column^[2]

The process observed in this experiment is a continuous process, therefore, there will be a constant feed stream entering the column. Also, as stated previously, the components of the feed stream with the higher boiling points will leave the column in a bottoms stream.

This experiment uses a binary system for the feed stream, therefore a McCabe-Thiele diagram analysis can be used. A McCabe-Thiele graph is formed by graphing the equilibrium data for the two components of the feed stream, the operating line for the rectifying section, the operating line for the stripping section, and a q line that is the mole fraction of the liquid. The rectifying section of the column is the section of the column that's above the location of the feed stream. The stripping section is the part of the column below the feed stream location. The number of theoretical stages required for the column to operate can then be determined. When constructing a McCabe-Thiele graph, the number of stages are counted by drawing steps from the operating lines to the equilibrium lines, as shown below. The location of the operating lines will have an effect on the number of theoretical stages. An example of a McCabe-Thiele diagram is shown below in Figure 2.

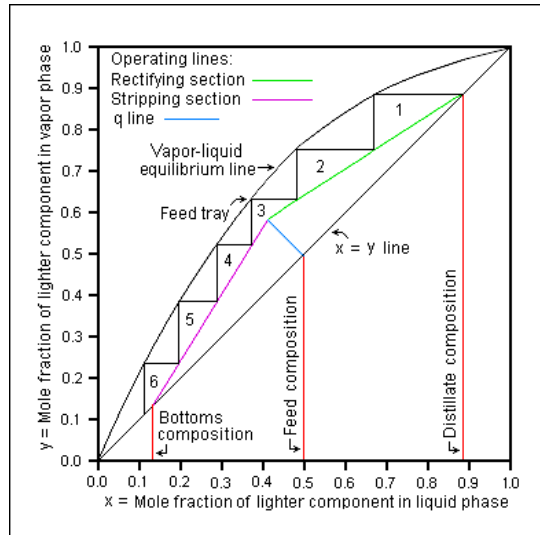


Figure 2: McCabe-Thiele Diagram^[5]

The Fenske equation can also be used to calculate the number of theoretical stages needed. The Fenske equation is as follows:

$$N = \frac{\log\left[\left(\frac{x_d}{1-x_d}\right)\left(\frac{1-x_b}{x_b}\right)\right]}{\log(\alpha)} \quad (2)$$

Where:

x_d is the mole fraction of the more volatile component in the distillate

x_b is the mole fraction of the more volatile component in the bottoms

α is the relative volatility of the mixture

The column efficiency can be determined using the following equation:

$$\eta = \frac{y_n - y_{n+1}}{y_n^* - y_{n+1}} \quad (3)$$

Where:

y_n^* is the composition of the vapor is in equilibrium with the composition of the liquid X_n .

Results

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Experiment A:

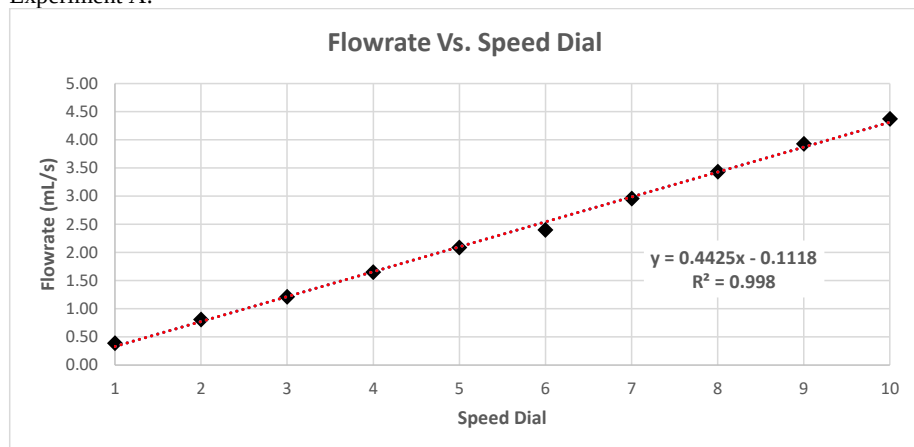


Figure 1: Flowrate Vs Speed of the pump

Figure 1 was obtained by changing the speed dial and measuring the flow rate of the liquid that was coming out of the feed tube. It can be seen that for each turn of the dial there was a fairly steady increase of the flowrate. The only outlier in the graph is at speed dial 6. Overall, there was an R^2 value of .998 which means that the pump is fairly accurate.

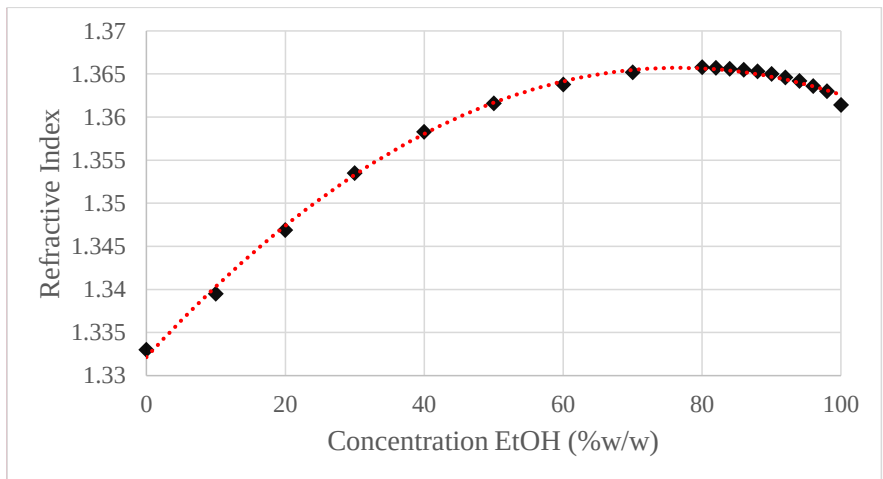
Experiment B:

During part B of the experiment the column was operating at a reflux ratio of 5:1. The feed was set to be positioned in the middle of the column and a feed stream at the bottom of the column. The following results were obtained

Table 3: Weight percent of samples in Experiment B

Experiment B			
Top Sample #	Refractive Index	%w/w EtOH	Goal RI
1	1.3624	99.19	1.3624
2	1.3623	99.36	1.3623
3	1.3628	98.48	1.3628
Experiment B			
Bot Sample #	Refractive Index	%w/w EtOH	Goal RI
1	1.3391	8.77	1.3391
2	1.3382	7.60	1.3382
3	1.3406	10.77	1.3406

Once reflux was started, a sample was taken from both the top and the bottom of the column. The refractive index was then taken in order to find how much ethanol was in the sample that was obtained. It can be seen as time progresses, the concentration of ethanol increases in the bottom sample while it decreases in the top sample. The concentration of ethanol was determined by taking the refractive index that was measured and plugging it in for y in the equation $y = -5.72 \cdot 10^{-6}x^2 + 8.766 \cdot 10^{-4}x + 1.332$. x, the concentration of ethanol, can then be solved for. A plot of the refractive index calibration curve is shown below.



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Graph 3: Calibration curve of the Concentration of Ethanol vs Refractive Index at 20°C.

The relative volatility of both ethanol and water can be calculated based on the concentration of ethanol that was obtained from the refractive index. The relative volatility was then used to calculate the theoretical plates which can be used to calculate the efficiency of the column. The following results were obtained.

Table 9: Volatility, Theoretical plates, and efficiency of the column

TOP RI		EtOH wt%		BOT RI		EtOH wt%	
1.36		98.48		1.34		10.77	
α (distillate)	α (Bottoms)	α (average)		N (Theoretical Plates)		η (Efficiency)	
0.32	8.14	1.60		13.29		1.66	

It can be seen within the data that there is a higher number of theoretical plates than there are actual plates. The number of actual plates is 8 while the theoretical is 13.

The temperature of the trays were recorded as the experiment progressed. It can be shown in the figure below that trays 1-4 operate at a lower temperature than the trays that are above them. It

can also be noticed that the split in temperature is exactly half way up the column and includes half of the plates. This makes sense considering the positions of the feed.

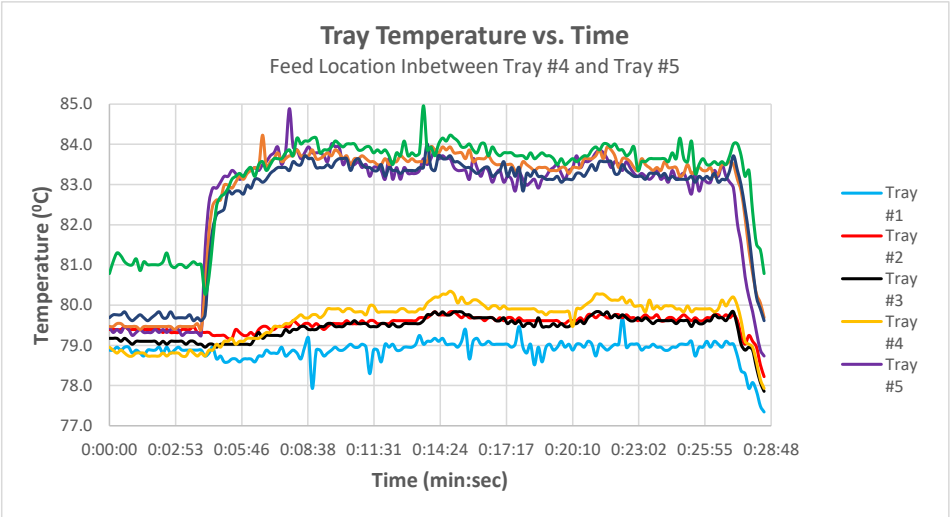


Figure 4: Temperature vs Time of Experiment 2

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Experiment C:

The power of the column was then set to .75 kW and allowed to equilibrate. Once the column was determined to be stable at total reflux, the bottom product flow, reflux stabilization, and the flow of the feed were started. Once the total reflux was determined to be stable, samples were taken from the top and bottom of the column to receive the results shown below.

Table 4: Results of Experiment C

Experiment C			
Top Sample #	Refractive Index	%w/w EtOH	Goal RI
1	1.3629	98.29	1.3629
2	1.3627	98.67	1.3627
3	1.3634	97.27	1.3634
Experiment C			
Bot Sample #	Refractive Index	%w/w EtOH	Goal RI
1	1.342	12.69	1.3420
2	1.343	14.10	1.3430
3	1.3441	15.69	1.3441

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It can be seen that the top sample decreased in ethanol as time progressed. In the bottom sample, the amount of ethanol increased as time went on and samples were taken. The refractive index was then used to calculate the relative volatility of the top and bottom stream. This was then used to calculate the theoretical plates to find the efficiency of the column.

The temperature throughout the experiment can be shown in the graph below. It was determined that the column was steady when there was bubbling all the way up the column, or when the temperature was constant. Due to the short amount of time and the column already being heated up, the temperature fluctuates up and down by around 3 degrees Celsius. This can be shown on the graph below.

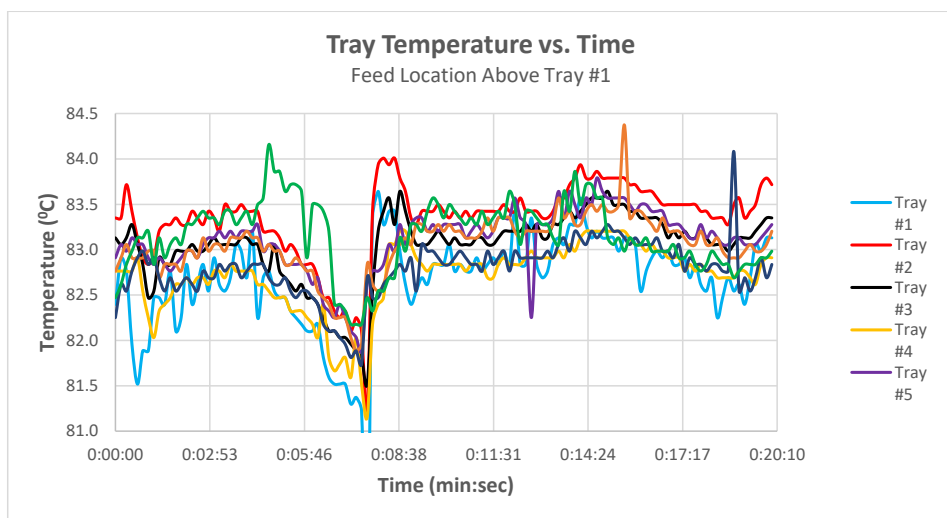


Figure 5: Temperature vs Time of Experiment 3

Discussion

Distillation columns are commonly used to separate out liquids by their boiling points. In this lab, a 10 wt% ethanol-water mixture was loaded into the reboiler of the column, and a 30 wt% ethanol-water mixture was used as a feed stream. Ethanol has a lower boiling point than water, so ethanol is expected to make up most of the top product.

The first part of the experiment was related to calibration of the peristaltic pump that would provide a feed stream to the column. The feed pump controls were limited to 0-10 dial, which limits the user when an exact flowrate is required. In order to make sure that the feed stream was accurate, the pump was calibrated by measuring flowrates against dial settings. As shown in figure 1, the flowrate and speed dial have a linear relationship. This was expected since the pump operates by a roller pushing on the feed tube. As more power is supplied to the pump with the dial, an equal response was shown by the flow rate.

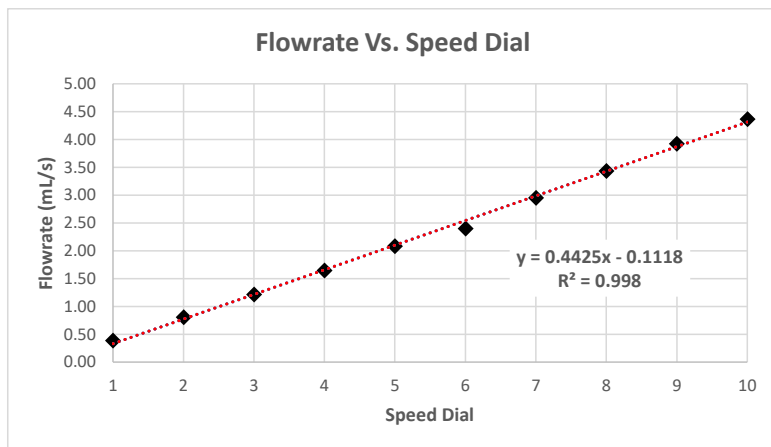


Figure 1. Flowrate vs Speed Dial

With the essential lab equipment calibrated, the next goal of the lab was to observe the composition of the top and bottom product. Ethanol composition was measured by comparing the sample's refractive index to known values as shown in figure 3.

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this discussion is almost the same as your results section. You put in the same graphs and table and you did not provide any explanation to your data.

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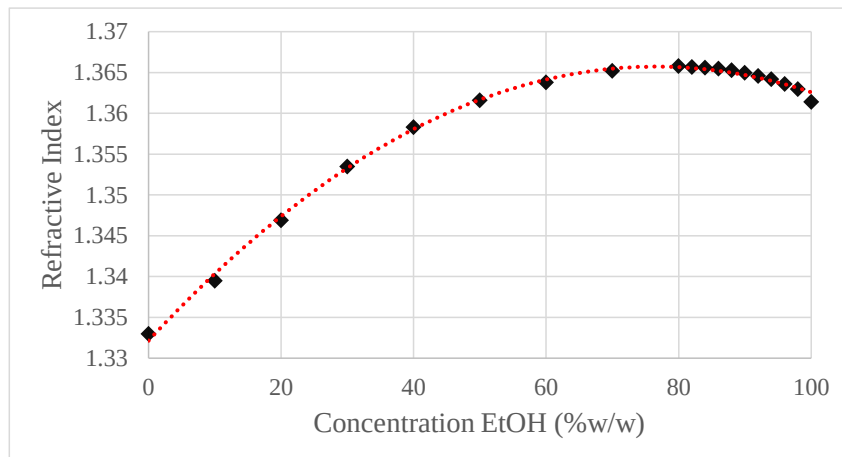


Figure 3. Calibration curve of Ethanol Refractive Index

For the second part of the lab, the feed was connected to the middle of the column. Since the feed was a different composition from the preloaded mixture, it was not allowed to flow until reflux was observed. If the feed valve had been opened early, the higher content ethanol mixture would cause inaccuracies in the bottom product as it would not have vaporized. In summary, the top and bottom product only were collected once the feed had started to flow into the middle of the column.

Table 3. Experiment B Ethanol Results

Experiment B			
Top Sample #	Refractive Index	%w/w EtOH	Goal RI
1	1.3624	99.19	1.3624
2	1.3623	99.36	1.3623
3	1.3628	98.48	1.3628
Experiment B			
Bot Sample #	Refractive Index	%w/w EtOH	Goal RI
1	1.3391	8.77	1.3391
2	1.3382	7.60	1.3382
3	1.3406	10.77	1.3406

Table 3 shows the wt% of Ethanol measured on both the top and bottom of the column. The trend of ethanol concentration increased in the top product and then flipped in the final sample. This is due to the fact that the additional ethanol added to the column would vaporize immediately. Once the column stabilized, the ethanol concentration increased at the bottom due

to the plate efficiency of inserting the feed at the middle of the column. The plate efficiency in part B was found to be 1.66, meaning the column had better separation than was expected. This is most likely due to there being a very large volume of mixture still preloaded into the reboiler, increasing the amount of ethanol.

The third part of the experiment was completely the same in procedure, however the feed was added to the top of the column instead of the middle. Table 4 shows the ethanol concentrations for part C. A similar trend with the concentrations was observed as that in part B. In this case the plate efficiency was found to be 0.94, so the column performed worse than it should have. This means that even though the column had 8 stages, the resulting separation data was as if it had only 7.5 stages. In the case of lower efficiency, it seems reasonable that the bottom product increased in concentration.

Table 3. Experiment C Ethanol Results

Experiment C			
Top Sample #	Refractive Index	%w/w EtOH	Goal RI
1	1.3629	98.29	1.3629
2	1.3627	98.67	1.3627
3	1.3634	97.27	1.3634
Experiment C			
Bot Sample #	Refractive Index	%w/w EtOH	Goal RI
1	1.342	12.69	1.3420
2	1.343	14.10	1.3430
3	1.3441	15.69	1.3441

When comparing both feed positions, the middle feed position had a much higher efficiency. The middle feed position was also better as the difference between top and bottom ethanol concentrations was higher. The higher the difference, the better the separation was in the column. Since the middle feed position had a higher difference, this further supports the efficiency results observed.

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References

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- [1] - Mujtaba, I.M. "Use of Continuous Distillation Columns for Batch Separation." *Chemical Engineering Research and Design*, Elsevier, 14 Feb. 2008, www.sciencedirect.com/science/article/pii/S0263876297715776.
- [2] - Continuous Distillation Lab Manual. Unit Operations Laboratory. 2020
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- [3] - Refractive index of ethanol solutions. refractometer.pl Web site.
<http://www.refractometer.pl/refraction-datasheet-ethanol>. Updated 2011. Accessed September 24th, 2019.
- [4] - "2.4 Binary Distillation." *Separation Process Engineering*, by Phillip C. Wankat, Prentice Hall, 2007, pp. 74–74.
- [5] - "File:McCabe-Thiele Diagram.svg." *File:McCabe-Thiele Diagram.svg*, 2008, commons.wikimedia.org/wiki/File:McCabe-Thiele_diagram.svg.

Appendix I: Data Tabulation

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Table 1 – Data collected from part A of the experiment. Shows the actual flow rates from the peristaltic pump at different speed dial settings.

Speed Dial	Volume (mL)	Time (s)	Flowrate (mL/s)	Flowrate (mL/min)
1	11.5	29.8	0.39	23.1
2	24.5	30.4	0.81	48.4
3	37.5	30.9	1.21	72.8
4	49.5	30.0	1.65	98.9
5	63.5	30.5	2.08	125.0
6	72.9	30.4	2.40	144.0
7	90.0	30.5	2.95	177.2
8	68.7	20.0	3.44	206.1
9	79.0	20.1	3.92	235.5
10	87.1	19.9	4.37	262.1

Table 2 – Values of the refractive index for various EtOH wt% aqueous solutions. Data was obtained from *refractometer.pl*. [3]

Concentration EtOH %w/w	0	10	20	30	40	50	60	70	80	82	84	86	88	90	92	94	96	98	100
n (20 °C)	1.333	1.3395	1.3469	1.3535	1.3583	1.3616	1.3638	1.3652	1.3658	1.3657	1.3656	1.3655	1.3653	1.365	1.3646	1.3642	1.3636	1.363	1.3614

Table 3 – Data collected from part B of the experiment. The refractive index of each sample is given along with the calculated wt% of EtOH in each sample. The ‘Goal RI’ is a manipulated variable used to determine the wt% of EtOH.

Experiment B			
Top Sample #	Refractive Index	%w/w EtOH	Goal RI
1	1.3624	99.19	1.3624
2	1.3623	99.36	1.3623
3	1.3628	98.48	1.3628
Experiment B			
Bot Sample #	Refractive Index	%w/w EtOH	Goal RI
1	1.3391	8.77	1.3391
2	1.3382	7.60	1.3382
3	1.3406	10.77	1.3406

Table 4 – Data collected from part C of the experiment. The refractive index of each sample is given along with the calculated wt% of EtOH in each sample. The ‘Goal RI’ is a manipulated variable used to determine the wt% of EtOH.

Experiment C			
Top Sample #	Refractive Index	%w/w EtOH	Goal RI
1	1.3629	98.29	1.3629
2	1.3627	98.67	1.3627
3	1.3634	97.27	1.3634
Experiment C			
Bot Sample #	Refractive Index	%w/w EtOH	Goal RI
1	1.342	12.69	1.3420
2	1.343	14.10	1.3430
3	1.3441	15.69	1.3441

Table 5 – Temperature data points for each tray and the reboiler for experiment B with the feed location in-between tray 4 and tray 5. *Note: this is a partial table of the data.*

Sample Number	Elapsed Time	Time	Temperature T1 (°C)	Temperature T2 (°C)	Temperature T3 (°C)	Temperature T4 (°C)	Temperature T5 (°C)	Temperature T6 (°C)	Temperature T7 (°C)	Temperature T8 (°C)	Temperature T9 (°C)
169	0:28:00	0:00:00	78.9	79.4	79.2	79.0	79.5	79.5	79.7	80.8	92.5
170	0:28:10	0:00:10	78.9	79.4	79.2	78.9	79.3	79.5	79.8	81.1	92.5
171	0:28:20	0:00:20	79.0	79.4	79.2	78.8	79.4	79.5	79.8	81.3	92.5
172	0:28:30	0:00:30	78.9	79.4	79.1	78.9	79.3	79.5	79.8	81.2	92.6
173	0:28:40	0:00:40	79.0	79.5	79.2	78.8	79.5	79.5	79.8	81.1	92.6
174	0:28:50	0:00:50	79.0	79.4	79.1	78.7	79.4	79.5	79.7	81.0	92.6
175	0:29:00	0:01:00	78.8	79.4	79.0	78.7	79.2	79.5	79.6	81.0	92.5
176	0:29:10	0:01:10	78.9	79.4	79.1	78.8	79.3	79.5	79.7	81.2	92.5
177	0:29:20	0:01:20	78.8	79.4	79.1	78.8	79.4	79.4	79.8	80.9	92.5
178	0:29:30	0:01:30	78.7	79.4	79.1	78.8	79.3	79.5	79.7	81.1	92.4
179	0:29:40	0:01:40	78.9	79.4	79.1	78.7	79.3	79.5	79.6	81.0	92.4
180	0:29:50	0:01:50	78.9	79.4	79.1	78.7	79.3	79.5	79.7	81.0	92.4
181	0:30:00	0:02:00	78.9	79.4	79.1	78.7	79.4	79.5	79.8	81.0	92.4
182	0:30:10	0:02:10	79.0	79.4	79.0	78.7	79.3	79.5	79.8	81.0	92.5
183	0:30:20	0:02:20	78.8	79.3	79.1	78.8	79.4	79.5	79.7	81.1	92.4
184	0:30:30	0:02:30	79.0	79.4	79.0	78.7	79.3	79.5	79.7	81.3	92.5
185	0:30:40	0:02:40	78.8	79.3	79.1	78.8	79.4	79.5	79.8	81.0	92.4

Table 6 – Temperature data points for each tray and the reboiler for experiment C with the feed location above tray 1. *Note: this is a partial table of the data.*

Sample Number	Elapsed Time	Time	Temperature T1 (°C)	Temperature T2 (°C)	Temperature T3 (°C)	Temperature T4 (°C)	Temperature T5 (°C)	Temperature T6 (°C)	Temperature T7 (°C)	Temperature T8 (°C)	Temperature T9 (°C)
426	1:10:50	0:00:00	82.4	83.3	83.1	82.8	82.9	82.8	82.3	82.5	91.0
427	1:11:00	0:00:10	82.9	83.3	83.1	82.8	83.1	82.9	82.6	82.6	91.0
428	1:11:10	0:00:20	82.8	83.7	83.1	82.8	83.0	83.1	82.5	82.8	91.0
429	1:11:20	0:00:30	82.0	83.4	83.3	82.8	83.1	82.9	82.8	83.0	91.0
430	1:11:30	0:00:40	81.5	83.1	83.1	82.9	83.1	82.9	82.9	83.1	90.9
431	1:11:40	0:00:50	81.9	82.9	82.8	82.5	83.1	82.9	83.0	83.1	91.0
432	1:11:50	0:01:00	81.9	82.8	82.5	82.3	82.9	83.0	82.8	83.2	90.9
433	1:12:00	0:01:10	82.5	82.9	82.5	82.0	82.8	83.0	82.9	82.8	91.0
434	1:12:10	0:01:20	82.5	83.2	82.9	82.3	82.9	82.7	82.7	83.1	91.0
435	1:12:20	0:01:30	82.4	83.3	82.8	82.4	82.8	82.8	82.5	82.9	91.0
436	1:12:30	0:01:40	82.8	83.3	82.8	82.5	82.8	82.8	82.7	83.1	90.9
437	1:12:40	0:01:50	82.1	83.3	83.0	82.6	82.8	82.8	82.5	83.1	91.0
438	1:12:50	0:02:00	82.3	83.3	83.0	82.6	82.9	82.8	82.6	83.1	90.9
439	1:13:00	0:02:10	82.8	83.3	83.0	82.7	83.0	83.0	82.7	83.3	90.7
440	1:13:10	0:02:20	82.4	83.4	83.1	82.6	83.0	83.0	82.6	83.3	90.9
441	1:13:20	0:02:30	82.6	83.3	83.0	82.7	82.9	82.9	82.5	83.3	91.0
442	1:13:30	0:02:40	82.8	83.3	83.0	82.7	83.1	83.1	82.8	83.4	90.8
443	1:13:40	0:02:50	82.5	83.4	83.1	82.6	83.1	83.0	82.7	83.3	90.8
444	1:13:50	0:03:00	82.9	83.3	83.1	82.8	83.1	83.1	82.7	83.3	90.9

Table 7 – Equilibrium data for EtOH and Water mixtures in mole fractions. Data was adapted from *Separations Process Engineering*. [1]

VLE Data (1 atm)		Operating Line	
x (mol)	y (mol)	x (mol)	y (mol)
0	0	0	0
0.019	0.17	0.019	0.019
0.0721	0.3891	0.0721	0.0721
0.0966	0.4375	0.0966	0.0966
0.1238	0.4704	0.1238	0.1238
0.1661	0.5089	0.1661	0.1661
0.2337	0.5445	0.2337	0.2337
0.2608	0.558	0.2608	0.2608
0.3273	0.5826	0.3273	0.3273
0.3965	0.6122	0.3965	0.3965
0.5198	0.6599	0.5198	0.5198
0.5732	0.6841	0.5732	0.5732
0.6763	0.7385	0.6763	0.6763
0.7472	0.7815	0.7472	0.7472
0.8943	0.8943	0.8943	0.8943
1	1	1	1

Table 8 – The Vapor Liquid Equilibrium (VLE) calculations for the vapor mole fraction of ethanol ('y') were determined by performing linear interpolation on the equilibrium data presented in Table 7. Where m is the slope, b is the intercept, x is the liquid mole fraction of ethanol, and y is the vapor mole fraction of ethanol as calculated by $y = m \cdot x + b$.

Vapor Fraction - Bot				Vapor Fraction - Top			
m	b	x	y	m	b	x	y
4.1262	0.0916	0.0451	0.2776	0.5276	0.3817	0.9621	0.8893

Vapor Fraction - Bot				Vapor Fraction - Top			
m	b	x	y	m	b	x	y
4.1262	0.0916	0.0678	0.3715	0.5276	0.3817	0.9330	0.8739

Table 9 – Calculation of the relative volatilities, number of theoretical trays, and column efficiency with different feed locations for experiment B (exp B) and experiment C (exp C).

	TOP RI	EtOH wt%	BOT RI	EtOH wt%	$\alpha_{\text{EtOH/H}_2\text{O}}$ (Top)	$\alpha_{\text{EtOH/H}_2\text{O}}$ (Bot)	$\langle \alpha_{\text{EtOH/H}_2\text{O}} \rangle$	N	η
EXP B	1.36	98.48	1.34	10.77	0.32	8.14	1.60	13.29	1.66
EXP C	1.36	97.27	1.34	15.69	0.50	8.12	2.01	7.52	0.94

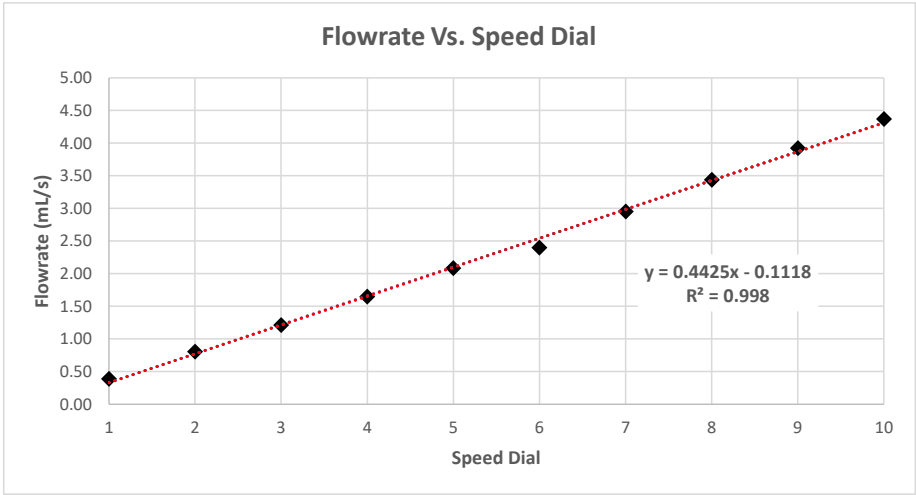


Figure 1: Accuracy of Flow rate Vs Speed dial

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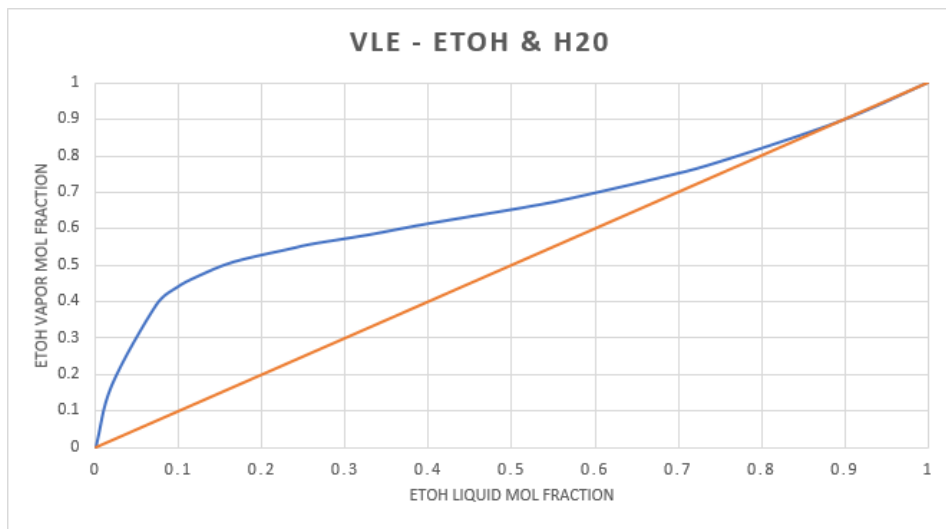


Figure 2: Mole fraction of ethanol vapor vs liquid

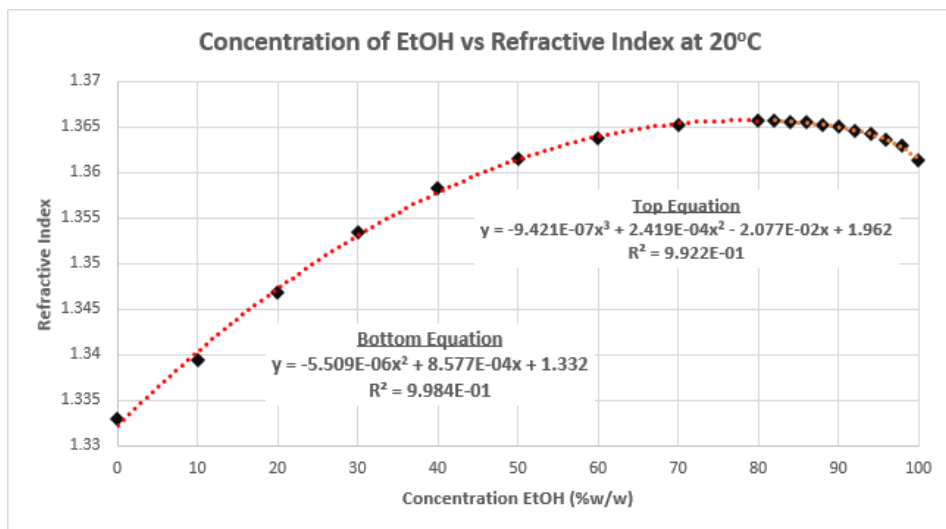


Figure 3: Concentration of ETOH vs Refractive Index

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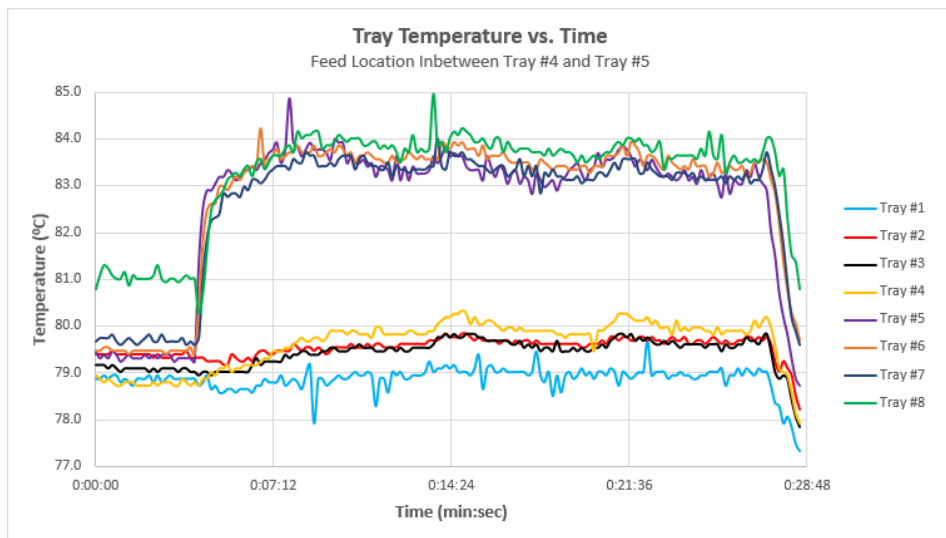


Figure 4: Tray temperature Vs temperature (Experiment B)

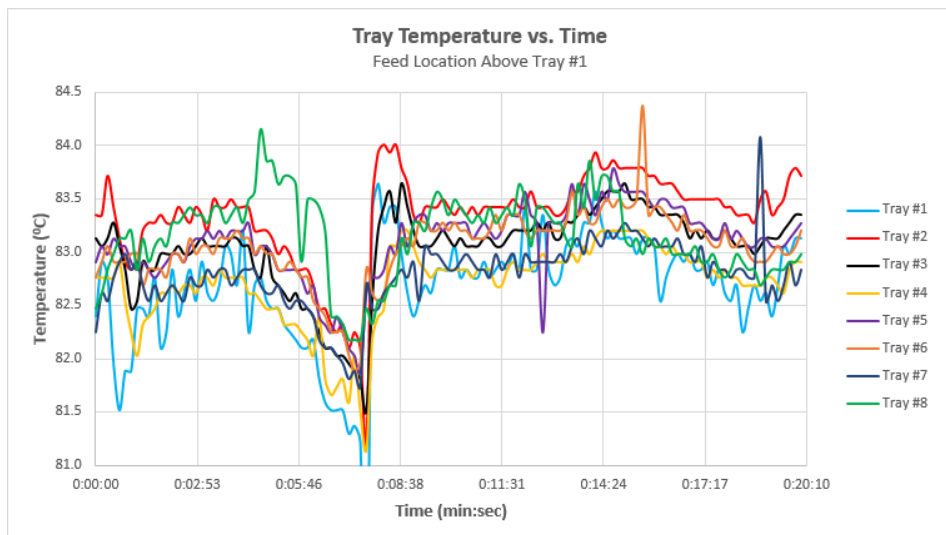


Figure 5: Tray temperature Vs temperature (Experiment C)

Appendix II: Error Analysis

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The errors in the continuous distillation lab that could impact our results are human errors in the reading of the analytical equipment for the pump calibration, errors in the method for using the refractometer, and not giving the column enough time to reach steady state. The 100 mL graduated cylinder used for calibrating the flow rate of the pump has an uncertainty of 1% and since you estimate the digit in-between the marked lines, human error is also involved. Also, there is a time delay between when the timer says stop and the person filling the graduated cylinder removes the tube with flowing solution from the graduated cylinder. We assumed that the total uncertainty of the measurement of the flow rate is $\pm 2\%$. The error in the weight percentages of the ethanol in the distillate and the bottoms was calculated by determining the mean and the standard deviation of the three samples. The standard deviation was calculated using the following equation:

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

Where σ is the standard deviation, x_i is the sample value, N is the sample size, and μ the average of samples refractive index. For part B, the distillate had a value of 99.01 ± 0.47 wt% and the bottoms had a value of 9.05 ± 1.6 wt%. For part C, the distillate had a value of 98.08 ± 0.73 wt% and the bottoms had a value of 14.16 ± 1.5 wt%. The errors in the wt% of ethanol for all the samples may be attributed to different sample volumes, insufficient cleaning of the refractometer, or insufficient cooling of the samples to a standard temperature before taking a reading. Also, since some of the values followed a trend as time went on, it is possible that the column did not reach steady state before we started taking samples. This would introduce error as we would have data that is not representative of the steady state conditions of the column.

Appendix III: Sample Calculations

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1. Determine α_{dist} and α_{bot}

$$\alpha_{dist} = \frac{(y_{EtOH}/x_{EtOH})}{(y_{H2O}/x_{H2O})}$$

Where α_{dist} is the relative volatility of ethanol and water in the distillate, y_i is the vapor mole composition, and x_i is the liquid mole composition.

$$\alpha_{dist} = 1.5133 = \frac{(0.8893/0.9621)}{((1 - 0.8893)/(1 - 0.9621))}$$

$$\alpha_{dist} = 0.32$$

The same equation is used to find the relative volatility in the bottoms of the column.

2. Determination of Average Relative Volatility $(\alpha_{EtOH/H2O})_{avg}$

$$(\alpha_{EtOH/H2O})_{avg} = \sqrt{\alpha_{dist} * \alpha_{bot}}$$

Where α_{dist} and α_{bot} are the relative volatilities of the distillate and the bottoms, respectively.

$$(\alpha_{EtOH/H2O})_{avg} = \sqrt{0.32 * 8.14} = 1.6$$

3. Determination of the Number of Theoretical Plates with Fenske Equation

$$N = \frac{\log \log \left[\left(\frac{x_{EtOH}}{x_{H2O}} \right)_{dist} \left(\frac{x_{H2O}}{x_{EtOH}} \right)_{bot} \right]}{\log \log (\alpha_{EtOH/H2O})_{avg}}$$

Where N is the number of theoretical plates.

$$N = 3.0587 = \frac{\log \left[\left(\frac{0.9621}{1 - 0.9621} \right)_{dist} \left(\frac{1 - 0.0451}{0.0451} \right)_{bot} \right]}{\log(1.60)}$$

$$N = 13.29$$

4. Determination of the Efficiency of the Column

$$E = \frac{\text{Number of Theoretical Plates}}{\text{Number of Actual Plates}} * 100\%$$

Where E is the efficiency of the column.

$$E = \frac{13.29}{8} * 100\% = 166\%$$

Appendix IV: Individual Contributions

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6.15	All
Pre-Lab Editing	1	All
Experimental Objective (Prelab)	0.5	Emily
Apparatus (prelab)	0.25	Peter/Allison
Materials & Supplies (prelab)	0.5	Peter/Allison
Procedure (prelab)	0.75	William / James
Project Safety Evaluation (prelab)	0.5	Emily / Diego
Final Lab Editing	1.5	All
Executive Summary (final lab)	1	Emily
Introduction (Final lab)	1	Diego
Theory (Final Lab)	3	Emily
Results (final lab)	4	Allison
Discussion (final lab)	4	Peter
References (final and/or prelab)	0.5	All
Data Tabulation/Graphs (final)	3	William / James
Error Analysis (final lab)	2	William / James
Sample Calculations (final Lab)	2	William / James
Power Point Presentation	2.5	All
Total	33.65	-